

# MCU (ESP32-S3): Working Document

Digital Audio Player • Rev 8

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## 1. Summary

| Field              | Value                                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------------|
| Manufacturer / MPN | Espressif Systems, ESP32-S3 (base chip)                                                             |
| Package            | 56-QFN, 7 × 7 mm, exposed pad (pin 57, GND)                                                         |
| Memory             | No in-package flash; no PSRAM (not used); external 8 MB quad SPI NOR flash, GigaDevice GD25B64EWIGR |
| Function           | Main controller: file decode, I2S to the DAC, UI, charger control, USB mass storage                 |
| Clock              | External 40 MHz passive crystal (compulsory), ECS-400-10-37B2-CKY-TR, $C_L = 10 \text{ pF}$         |

Source: [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[11\]](#)

## 2. Design Requirements

| Requirement | Value                                     | Source              |
|-------------|-------------------------------------------|---------------------|
| Supply      | +3V3_DIG; 3.0 to 3.6 V on all supply pins | <a href="#">[2]</a> |

| Requirement | Value                                      | Source      |
|-------------|--------------------------------------------|-------------|
| Crystal     | 40 MHz only; accuracy within $\pm 10$ ppm  | [2]         |
| Storage     | microSD over 1-bit SDMMC, card detect      | Author      |
| Audio       | I2S slave to the ES9038Q2M (DAC is master) | Author      |
| Control     | I2C to the DAC and the BQ24250 at 100 kHz  | Author      |
| Display     | 4-wire SPI to the ER-TFT024IPS-3           | Author      |
| USB         | USB-C, mass storage (GPIO19 / GPIO20)      | Author, [2] |
| Constraints | Low noise · Small footprint                | Author      |

## 3. Pin Configuration

### 3.1 IO Pin Default Configuration

| No. | Name       | Power      | At Reset | After Reset |
|-----|------------|------------|----------|-------------|
| 1   | LNA_IN     |            |          |             |
| 2   | VDD3P3     |            |          |             |
| 3   | VDD3P3     |            |          |             |
| 4   | CHIP_PU    | VDD3P3_RTC |          |             |
| 5   | GPIO0      | VDD3P3_RTC | IE, WPU  | IE, WPU     |
| 6   | GPIO1      | VDD3P3_RTC | IE       | IE          |
| 7   | GPIO2      | VDD3P3_RTC | IE       | IE          |
| 8   | GPIO3      | VDD3P3_RTC | IE       | IE          |
| 9   | GPIO4      | VDD3P3_RTC |          |             |
| 10  | GPIO5      | VDD3P3_RTC |          |             |
| 11  | GPIO6      | VDD3P3_RTC |          |             |
| 12  | GPIO7      | VDD3P3_RTC |          |             |
| 13  | GPIO8      | VDD3P3_RTC |          |             |
| 14  | GPIO9      | VDD3P3_RTC |          | IE          |
| 15  | GPIO10     | VDD3P3_RTC |          | IE          |
| 16  | GPIO11     | VDD3P3_RTC |          | IE          |
| 17  | GPIO12     | VDD3P3_RTC |          | IE          |
| 18  | GPIO13     | VDD3P3_RTC |          | IE          |
| 19  | GPIO14     | VDD3P3_RTC |          | IE          |
| 20  | VDD3P3_RTC |            |          |             |
| 21  | XTAL_32K_P | VDD3P3_RTC |          |             |
| 22  | XTAL_32K_N | VDD3P3_RTC |          |             |
| 23  | GPIO17     | VDD3P3_RTC |          | IE          |
| 24  | GPIO18     | VDD3P3_RTC |          | IE          |
| 25  | GPIO19     | VDD3P3_RTC |          |             |
| 26  | GPIO20     | VDD3P3_RTC | USB_PU   | USB_PU      |

| No. | Name       | Power                | At Reset | After Reset |
|-----|------------|----------------------|----------|-------------|
| 27  | GPIO21     | VDD3P3_RTC           |          |             |
| 28  | SPICS1     | VDD_SPI              | IE, WPU  | IE, WPU     |
| 29  | VDD_SPI    |                      |          |             |
| 30  | SPIHD      | VDD_SPI              | IE, WPU  | IE, WPU     |
| 31  | SPIWP      | VDD_SPI              | IE, WPU  | IE, WPU     |
| 32  | SPICS0     | VDD_SPI              | IE, WPU  | IE, WPU     |
| 33  | SPICLK     | VDD_SPI              | IE, WPU  | IE, WPU     |
| 34  | SPIQ       | VDD_SPI              | IE, WPU  | IE, WPU     |
| 35  | SPID       | VDD_SPI              | IE, WPU  | IE, WPU     |
| 36  | SPICLK_N   | VDD_SPI / VDD3P3_CPU | IE       | IE          |
| 37  | SPICLK_P   | VDD_SPI / VDD3P3_CPU | IE       | IE          |
| 38  | GPIO33     | VDD_SPI / VDD3P3_CPU |          | IE          |
| 39  | GPIO34     | VDD_SPI / VDD3P3_CPU |          | IE          |
| 40  | GPIO35     | VDD_SPI / VDD3P3_CPU |          | IE          |
| 41  | GPIO36     | VDD_SPI / VDD3P3_CPU |          | IE          |
| 42  | GPIO37     | VDD_SPI / VDD3P3_CPU |          | IE          |
| 43  | GPIO38     | VDD3P3_CPU           |          | IE          |
| 44  | MTCK       | VDD3P3_CPU           |          | IE          |
| 45  | MTDO       | VDD3P3_CPU           |          | IE          |
| 46  | VDD3P3_CPU |                      |          |             |
| 47  | MTDI       | VDD3P3_CPU           |          | IE          |
| 48  | MTMS       | VDD3P3_CPU           |          | IE          |
| 49  | U0TXD      | VDD3P3_CPU           | IE, WPU  | IE, WPU     |
| 50  | U0RXD      | VDD3P3_CPU           | IE, WPU  | IE, WPU     |
| 51  | GPIO45     | VDD3P3_CPU           | IE, WPD  | IE, WPD     |
| 52  | GPIO46     | VDD3P3_CPU           | IE, WPD  | IE, WPD     |
| 53  | XTAL_N     |                      |          |             |
| 54  | XTAL_P     |                      |          |             |
| 55  | VDDA       |                      |          |             |
| 56  | VDDA       |                      |          |             |
| 57  | GND        |                      |          |             |

IE: input enabled. WPU: internal weak pull-up enabled. WPD: internal weak pull-down enabled. USB\_PU: USB pull-up resistor enabled.  
Source: [\[2\]](#), section GPIO.

### 3.2 Power-Up Glitches on Pins

| Pin        | Glitch <sup>a</sup>                               | Typical Time (μs) |
|------------|---------------------------------------------------|-------------------|
| GPIO1      | Low-level glitch                                  | 60                |
| GPIO2      | Low-level glitch                                  | 60                |
| GPIO3      | Low-level glitch                                  | 60                |
| GPIO4      | Low-level glitch                                  | 60                |
| GPIO5      | Low-level glitch                                  | 60                |
| GPIO6      | Low-level glitch                                  | 60                |
| GPIO7      | Low-level glitch                                  | 60                |
| GPIO8      | Low-level glitch                                  | 60                |
| GPIO9      | Low-level glitch                                  | 60                |
| GPIO10     | Low-level glitch                                  | 60                |
| GPIO11     | Low-level glitch                                  | 60                |
| GPIO12     | Low-level glitch                                  | 60                |
| GPIO13     | Low-level glitch                                  | 60                |
| GPIO14     | Low-level glitch                                  | 60                |
| XTAL_32K_P | Low-level glitch                                  | 60                |
| XTAL_32K_N | Low-level glitch                                  | 60                |
| GPIO17     | Low-level glitch                                  | 60                |
| GPIO18     | Low-level / High-level glitch                     | 60                |
| GPIO19     | Low-level glitch / High-level glitch <sup>b</sup> | 60                |
| GPIO20     | Pull-down glitch / High-level glitch <sup>b</sup> | 60                |

<sup>a</sup> Low-level: pin outputs low for the period. High-level: pin outputs high for the period. Pull-down: pin is held by the internal weak pull-down for the period. <sup>b</sup> GPIO19 and GPIO20 each show two high-level glitches of about 60 μs during power-up; total duration and delay are 3.2 ms (GPIO19) and 2 ms (GPIO20).

Source: [2], section GPIO, table Power-Up Glitches on Pins.

### 3.3 GPIO Assignment

Current connection of every GPIO-capable pin, read from the schematic sheets. X marks a recommended change; the comment ID points to the table that follows. Reset state is at reset / after reset, from Section 3.1 (Hi-Z: no input, no pull). Sheets: MCU-3, Audio-1, Power-2, Periph-4.

| Pin | Name  | Net / connection                   | Sheet            | Dir | Reset   | External pull | Flags                | Sw. | Cmt. |
|-----|-------|------------------------------------|------------------|-----|---------|---------------|----------------------|-----|------|
| 5   | GPIO0 | BOOT0: BOOT switch to GND via 1 kΩ | MCU-3            | In  | IE, WPU | 10 kΩ pu      | Strap (boot)         |     | C1   |
| 6   | GPIO1 | GPIO1_DAC                          | Audio-1          | I/O | IE      | 10 kΩ pd      | Glitch               |     | C2   |
| 7   | GPIO2 | GPIO2_DAC                          | Audio-1          | I/O | IE      | 10 kΩ pd      | Glitch               |     | C2   |
| 8   | GPIO3 | TRI_STATE_24                       | Audio-1          | Out | IE      | 10 kΩ pd (R8) | Strap (JTAG); glitch |     | C3   |
| 9   | GPIO4 | SDA_DAC / SDA_BQ                   | Audio-1, Power-2 | I/O | Hi-Z    | 10 kΩ pu × 2  | Glitch               |     | C4   |
| 10  | GPIO5 | SCL_DAC / SCL_BQ                   | Audio-1, Power-2 | Out | Hi-Z    | 10 kΩ pu × 2  | Glitch               |     | C4   |
| 11  | GPIO6 | ADDR_DAC                           | Audio-1          | Out | Hi-Z    | 10 kΩ pu (R9) | Glitch               |     |      |
| 12  | GPIO7 | DATA_CLK_DAC (I2S bit clock)       | Audio-1          | In  | Hi-Z    | None          | Glitch               |     |      |

| Pin | Name                | Net / connection                          | Sheet    | Dir | Reset     | External pull                       | Flags               | Sw. | Cmt.          |
|-----|---------------------|-------------------------------------------|----------|-----|-----------|-------------------------------------|---------------------|-----|---------------|
| 13  | GPIO8               | DATA1_DAC (I2S frame clock)               | Audio-1  | In  | Hi-Z      | None                                | Glitch              |     |               |
| 14  | GPIO9               | DATA2_DAC (I2S data)                      | Audio-1  | Out | Hi-Z / IE | None                                | Glitch              |     |               |
| 15  | GPIO10              | PG_+3V3                                   | Power-2  | In  | Hi-Z / IE | 10 kΩ pu (R224)                     | Glitch              |     |               |
| 16  | GPIO11              | PG_-3V3                                   | Power-2  | In  | Hi-Z / IE | 10 kΩ pu (R222)                     | Glitch              | X   | C16           |
| 17  | GPIO12              | SELECT (encoder push switch)              | Periph-4 | In  | Hi-Z / IE | 10 kΩ pd                            | Glitch; RTC         | X   | C12, C17, C18 |
| 18  | GPIO13              | TRI_STATE_22                              | Audio-1  | Out | Hi-Z / IE | 10 kΩ pd (R7)                       | Glitch              |     |               |
| 19  | GPIO14              | VOLUME+                                   | Periph-4 | In  | Hi-Z / IE | 10 kΩ pd                            | Glitch; RTC         |     | C12, C17      |
| 21  | GPIO15 (XTAL_32K_P) | ENC_B                                     | Periph-4 | In  | Hi-Z      | 10 kΩ pu, 10 kΩ series, 10 nF       | Glitch; RTC         |     | C5            |
| 22  | GPIO16 (XTAL_32K_N) | ENC_A                                     | Periph-4 | In  | Hi-Z      | 10 kΩ pu, 10 kΩ series, 10 nF       | Glitch; RTC         |     | C5            |
| 23  | GPIO17              | VOLUME-                                   | Periph-4 | In  | Hi-Z / IE | 10 kΩ pd                            | Glitch; RTC         |     | C12, C17      |
| 24  | GPIO18              | RESET_LCD                                 | Periph-4 | Out | Hi-Z / IE | None                                | Glitch (low / high) |     | C8            |
| 25  | GPIO19              | DN (USB D-)                               | Power-2  | I/O | Hi-Z      | 22 Ω series                         | Glitch; USB         |     |               |
| 26  | GPIO20              | DP (USB D+)                               | Power-2  | I/O | USB_PU    | 22 Ω series                         | Glitch; USB         |     |               |
| 27  | GPIO21              | POWER                                     | Periph-4 | In  | Hi-Z      | 10 kΩ pd                            | RTC (wake)          |     | C12           |
| 28  | SPICS1 (GPIO26)     | CD_SD (card detect)                       | Periph-4 | In  | IE, WPU   | 10 kΩ pu                            |                     |     | C6            |
| 30  | SPIHD (GPIO27)      | SPI_HD; flash pin 7 (HOLD# / IO3)         | MCU-3    | I/O | IE, WPU   |                                     | Flash               |     | C9            |
| 31  | SPIWP (GPIO28)      | SPI_WP; flash pin 3 (WP# / IO2)           | MCU-3    | I/O | IE, WPU   |                                     | Flash               |     | C9            |
| 32  | SPICS0 (GPIO29)     | SPI_CS0; flash pin 1 (CS#)                | MCU-3    | Out | IE, WPU   |                                     | Flash               |     | C9            |
| 33  | SPICLK (GPIO30)     | SPICLK; 22 Ω series to flash pin 6 (SCLK) | MCU-3    | Out | IE, WPU   |                                     | Flash               |     | C9            |
| 34  | SPIQ (GPIO31)       | SPI_Q; flash pin 2 (SO / IO1)             | MCU-3    | I/O | IE, WPU   |                                     | Flash               |     | C9            |
| 35  | SPID (GPIO32)       | SPI_D; flash pin 5 (SI / IO0)             | MCU-3    | I/O | IE, WPU   |                                     | Flash               |     | C9            |
| 36  | SPICLK_N (GPIO48)   | NC; reserved BL_PWM                       |          | Out | IE        | None                                | VDD_SPI domain      | X   | C7, C15       |
| 37  | SPICLK_P (GPIO47)   | CLK_SD; 22 Ω series                       | Periph-4 | Out | IE        | None                                | VDD_SPI domain      |     | C6            |
| 38  | GPIO33              | PLUG_DET                                  | Audio-1  | In  | Hi-Z / IE | 100 kΩ pu (R19), 10 kΩ series (R20) |                     |     |               |
| 39  | GPIO34              | EN_SPEAKER                                | Audio-1  | Out | Hi-Z / IE | 50 kΩ pd (R31) via 54.9 kΩ (R29)    |                     |     |               |
| 40  | GPIO35              | SHDN_BB_3.9                               | Power-2  | Out | Hi-Z / IE | 1 MΩ pd (MCU-3)                     |                     |     | C10           |
| 41  | GPIO36              | RESETB_DAC                                | Audio-1  | Out | Hi-Z / IE | 10 kΩ pd (R12)                      |                     |     | C2            |
| 42  | GPIO37              | INT_BQ                                    | Power-2  | In  | Hi-Z / IE | 10 kΩ pu                            |                     |     |               |
| 43  | GPIO38              | D/CX_LCD                                  | Periph-4 | Out | Hi-Z / IE | None                                |                     |     |               |
| 44  | MTCK (GPIO39)       | SCL_LCD; 22 Ω series                      | Periph-4 | Out | Hi-Z / IE | None                                | JTAG pin            |     |               |
| 45  | MTDO (GPIO40)       | MOSI_LCD                                  | Periph-4 | Out | Hi-Z / IE | None                                | JTAG pin            |     |               |
| 47  | MTDI (GPIO41)       | CMD_SD                                    | Periph-4 | I/O | Hi-Z / IE | 10 kΩ pu                            | JTAG pin            |     | C6            |
| 48  | MTMS (GPIO42)       | DAT0_SD                                   | Periph-4 | I/O | Hi-Z / IE | 10 kΩ pu                            | JTAG pin            |     | C6            |
| 49  | U0TXD (GPIO43)      | DOWN; 1 kΩ series                         | Periph-4 | In  | IE, WPU   | 10 kΩ pd                            | UART0 TX            |     | C11, C12      |
| 50  | U0RXD (GPIO44)      | UP                                        | Periph-4 | In  | IE, WPU   | 10 kΩ pd                            | UART0 RX            |     | C11, C12      |
| 51  | GPIO45              | CS_LCD                                    | Periph-4 | Out | IE, WPD   | 10 kΩ pd                            | Strap (VDD_SPI)     |     | C13           |
| 52  | GPIO46              | CE_BQ                                     | Power-2  | Out | IE, WPD   | Internal WPD                        | Strap (boot)        |     | C14           |

Source: [8]; reset states and flags from [2]; strapping from [3].

| ID  | Comment                                                                                                                                                                                                                                                                                                                  | Source               |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| C1  | Fitted: 10 kΩ pull-up on BOOT0, BOOT switch to GND through 1 kΩ ( $\approx 0.3$ V when pressed), RESET switch from CHIP_PU to GND. Download mode: hold BOOT, press and release RESET, release BOOT.                                                                                                                      | [2], [3], [8], [9]   |
| C2  | 10 kΩ pull-downs on GPIO1_DAC, GPIO2_DAC and RESETB_DAC confirmed: the DAC GPIO default mode shuts the DAC down when high.                                                                                                                                                                                               | [15]                 |
| C3  | GPIO3 now carries TRI_STATE_24. Its strap selects the JTAG source only when the eFuse enabling it is burned; R8 gives a defined low at reset, and the power-up low glitch keeps the oscillator disabled.                                                                                                                 | [3], [8]             |
| C4  | Two 10 kΩ pull-up pairs on the I2C bus (DAC sheet R10 / R11, charger sheet): 5 kΩ effective. Acceptable at 100 kHz; one pair is redundant.                                                                                                                                                                               | [8]                  |
| C5  | Encoder (Alps EC12D1564404): ENC_A on GPIO16, ENC_B on GPIO15, both RTC-capable (wake). Per line: 10 kΩ pull-up, 10 kΩ series, 10 nF to GND on the GPIO side; ESD array on ENC_A, ENC_B and SELECT.                                                                                                                      | [8], [9], [20]       |
| C6  | microSD (Same Sky MSD-1-A), 1-bit SDMMC: CLK_SD on GPIO47 (22 Ω series at the MCU), CMD_SD on GPIO41, DAT0_SD on GPIO42, CD_SD on GPIO26 (10 kΩ pull-up; low with a card inserted). 10 kΩ pull-ups on CMD and DAT0 to DAT3; TVS array on CLK, CMD, DAT0 and DAT3. 4-bit not used (would need GPIO1, GPIO2 and ADDR_DAC). | [8], [9], [19]       |
| C7  | GPIO48 reserved for the backlight PWM; the drive circuit is open in Peripherals Rev 9. GPIO47 / GPIO48 sit in the VDD_SPI domain and are 3.3 V only while GPIO45 = 0 and the flash is 3.3 V.                                                                                                                             | [2], [9]             |
| C8  | Power-up glitch pulses the LCD reset; harmless because firmware resets the panel at start-up.                                                                                                                                                                                                                            | [2]                  |
| C9  | Flash connected per the GD25Q64E 8-WSON pinout; VCC on VDD_SPI with 0.1 μF and 1 μF; pins 4 and 9 to GND. SPIWP / SPIHD / SPICSO internal pull-ups hold WP#, HOLD# and CS# inactive at reset. SPICLK: 22 Ω series fitted (place at pin 33); an unpopulated shunt-C footprint is optional per the SPI EMC note.           | [2], [8], [11], [13] |
| C10 | 1 MΩ pull-down (#1-123) fitted on the GPIO side of R203, per the LTC3440 datasheet; holds the analog rails off until firmware enables them.                                                                                                                                                                              | [10], [8], [12]      |
| C11 | The ROM drives U0TXD during boot. A press while TX is low shorts the driver to +3V3 through the switch; a 1 kΩ series resistor (fitted) limits this to 3.3 mA. See FW5. GPIO43 / GPIO44 are not RTC-capable: UP / DOWN cannot wake the player.                                                                           | [2], [3], [8]        |
| C12 | Buttons are active high (10 kΩ pull-down, switch to +3V3_DIG); Peripherals Rev 9 updated to match. GPIO inputs are high impedance, so either polarity works. No hardware debounce (FW6).                                                                                                                                 | [8], [9]             |
| C13 | Keep the pull-down. A pull-up (usual for CS) makes GPIO45 = 1 at reset and selects 1.8 V on VDD_SPI.                                                                                                                                                                                                                     | [2]                  |
| C14 | CE low at reset keeps charging enabled. GPIO46 = 1 with GPIO0 = 0 is an invalid strap combination.                                                                                                                                                                                                                       | [3]                  |
| C15 | All GPIOs assigned except GPIO48 (backlight PWM, reserved).                                                                                                                                                                                                                                                              | [8]                  |
| C16 | LT3093 PG low is specified at 100 μA; the 10 kΩ pull-up (R222) sinks $\approx 330$ μA, so a valid low is not guaranteed. Raise R222 to 33 to 47 kΩ or confirm V(PG low) at 330 μA.                                                                                                                                       | [17]                 |
| C17 | SELECT (GPIO12), VOLUME+ (GPIO14) and VOLUME- (GPIO17) output a 60 μs low glitch at power-up. A button held during power-up shorts that pin to +3V3 through the switch for the glitch period. Low risk; 1 kΩ in series, as on DOWN, removes it.                                                                          | [2], [8]             |
| C18 | As drawn, the Interface Buttons block ties the SELECT node to the +3V3_DIG button bus (wire from the bus to the SELECT pull-down node), so GPIO12 reads high permanently. Remove that wire; the encoder switch (D to +3V3_DIG, E to SELECT) is the only switch on this net.                                              | [8]                  |

## 4. External Components

| Ref              | AL-PN  | Part                               | Notes                                                                                                                                                                                                         | Qty |
|------------------|--------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| U                |        | Espressif ESP32-S3                 | 56-QFN, 7 × 7 mm                                                                                                                                                                                              | 1   |
| Y                | #1-118 | ECS Inc.<br>ECS-400-10-37B2-CKY-TR | 40 MHz crystal, $C_L$ 10 pF, ±10 ppm tolerance, ±10 ppm stability, ESR 40 Ω, 4-SMD. DigiKey 50-ECS-400-10-37B2-CKY-TRCT-ND. [4], [5]                                                                          | 1   |
| C (XTAL)         | #1-122 | Murata<br>GJM1555C1H120FB01D       | 12 pF, ±1 %, 50 V, COG, 0402. Crystal load capacitors, one per crystal pin (C1 / C4 in the guideline figure). DigiKey 490-11190-1-ND. [6]                                                                     | 2   |
| L (XTAL_P)       | #1-120 | TDK MHQ1005075HA24N<br>HTD25       | 24 nH, ±3 %, 250 mA, 700 mΩ max, 0402. Series component on XTAL_P; 24 nH is the guideline starting value. DigiKey 445-MHQ1005075HA24NHTD25CT-ND. [2], [7]                                                     | 1   |
| L (VDD3P3)       | #1-121 | TDK MHQ1005075HA2N0<br>BTD25       | 2 nH, 1 A, 50 mΩ max, 0402. LC filter inductor on the VDD3P3 rail (pins 2 and 3); guideline asks for ≥ 500 mA rating. DigiKey 445-MHQ1005075HA2N0BTD25CT-ND. [2], [7]                                         | 1   |
| U (flash)        | #1-124 | GigaDevice<br>GD25B64EWIGR         | 64 Mbit (8 MB) SPI NOR flash, quad I/O, 133 MHz, 2.7 to 3.6 V, -40 to 85 °C, 8-WSON (6 × 5 mm), exposed pad. Powered from VDD_SPI. Footprint SON127P600X500X80-9N. DigiKey 1970-GD25B64EWIGRCT-ND. [11], [13] | 1   |
| R (GPIO35)       | #1-123 | YAGEO RE0402BRE071ML               | 1 MΩ, ±0.1 %, 1/16 W, 0402, thick film. SHDN_BB_3.9 pull-down on the GPIO side of R203. DigiKey 13-RE0402BRE071MLCT-ND. [10], [12]                                                                            | 1   |
| SW (BOOT, RESET) |        | C&K; PTS810 SJM 250<br>SMTR LFS    | Top-actuated SMD tactile, 4.2 × 3.2 mm, same part as the front switches. BOOT on GPIO0, RESET on CHIP_PU; internal, used with the enclosure open. [9]                                                         | 2   |

## 5. Equations and Results

Crystal load capacitors (C1 = C4)

$$C_L = \frac{C_1 C_4}{C_1 + C_4} + C_{\text{stray}} \quad (1)$$

$$C_1 = C_4 \Rightarrow C_1 = C_4 = 2(C_L - C_{\text{stray}}) \quad (2)$$

$$C_{L, \text{eff}} = \frac{C_1}{2} + C_{\text{stray}} \quad (3)$$

Source: equation (1) and C1 / C4 naming per [2], section External Crystal Clock Source. Equations (2) and (3) follow from (1) with C1 = C4.

| Parameter                                  | Value                                             | Source   |
|--------------------------------------------|---------------------------------------------------|----------|
| $C_L$ (crystal load capacitance)           | 10 pF                                             | [4], [5] |
| $C_{\text{stray}}$ (PCB stray capacitance) | 3 to 5 pF (assumed; the guideline gives no value) | Estimate |
| C1 = C4 required                           | 10 to 14 pF                                       | Eq. (2)  |
| C1 = C4 selected                           | 12 pF (#1-122), centre of the required range      | Eq. (2)  |

| Parameter              | Value                                                                                                                                                                      | Source  |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $C_{L,eff}$ with 12 pF | 9 to 11 pF ( $C_L = 10 \text{ pF} \pm 1 \text{ pF}$ )                                                                                                                      | Eq. (3) |
| Tuning                 | Set after assembly: TX tone test, measure 2.4 GHz frequency offset, trim to within $\pm 10$ ppm. Positive offset: increase load capacitance. Negative offset: decrease it. | [2]     |

#### Flash package and land pattern (8-WSON, $6 \times 5 \text{ mm}$ )

| Symbol | Meaning                              | Min  | Nom  | Max  |
|--------|--------------------------------------|------|------|------|
| D      | Body length (lead to lead direction) | 5.90 | 6.00 | 6.10 |
| E      | Body width (pitch direction)         | 4.90 | 5.00 | 5.10 |
| A      | Height                               | 0.70 | 0.75 | 0.80 |
| e      | Pitch                                |      | 1.27 |      |
| b      | Lead width                           | 0.35 | 0.40 | 0.50 |
| L      | Lead length                          | 0.50 | 0.60 | 0.75 |
| D2     | Exposed pad along D                  | 3.30 | 3.40 | 3.50 |
| E2     | Exposed pad along E                  | 3.90 | 4.00 | 4.10 |

All dimensions mm. Source: [13], Section 10.6 (GD25Q64E; same GigaDevice WSON8  $6 \times 5 \text{ mm}$  outline, not yet confirmed in a GD25B64E datasheet). Exposed pad is floating.

| Land pattern item           | Value                                                                                                             | Source           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------|------------------|
| IPC-7351 name               | SON127P600X500X80-9N (library #1-124)                                                                             | Author           |
| Pin order                   | Counter-clockwise from pin 1 viewed from the top: 1 to 4 down one side, 5 to 8 up the other; pin 8 opposite pin 1 | [13]             |
| Pinout                      | 1 CS#, 2 SO / IO1, 3 WP# / IO2, 4 VSS, 5 SI / IO0, 6 SCLK, 7 HOLD# / IO3, 8 VCC, 9 exposed pad                    | [13]             |
| Lead lands (reviewed)       | $\approx 1.01 \times 0.39 \text{ mm}$ , spanning 2.49 to 3.50 mm from centre                                      | Author footprint |
| Exposed pad land (reviewed) | $\approx 3.50 \times 3.97 \text{ mm}$ ; $\approx 0.74 \text{ mm}$ to the lead lands                               | Author footprint |
| Exposed pad paste           | Split into windowpanes, $\approx 50 \%$ coverage                                                                  | Recommendation   |

## 6. Guideline Notes

| Topic     | Note                                                                                                                                                                                                                   | Source                   |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| SPI (EMC) | Place a series resistor or ferrite bead plus a capacitor to ground on the SPI_CLK trace. Where space allows, add the same series R and shunt C on the other SPI traces. Keep the RC / LC parts close to the chip pins. | [2], SPI                 |
| XTAL_P    | Series component on XTAL_P, 24 nH initial value, adjusted after an overall test, to limit crystal harmonics reaching the RF path.                                                                                      | [2], Clock Source        |
| VDD3P3    | LC filter on the VDD3P3 rail to suppress high-frequency harmonics; inductor rated 500 mA or more. 10 $\mu\text{F}$ on the rail recommended because TX bursts can collapse it.                                          | [2], Analog Power Supply |
| Crystal   | Crystal amplitude recommended above 500 mV. Frequency deviation beyond $\pm 10$ ppm degrades RF performance.                                                                                                           | [2], Clock Source        |

| Topic     | Note                                                                                                                                                                                                           | Source                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| CHIP_PU   | RC delay recommended, typically $R = 10\text{ k}\Omega$ and $C = 1\text{ }\mu\text{F}$ ; must not float.                                                                                                       | [2], Power-up and Reset |
| Strapping | GPIO0, GPIO3, GPIO45 and GPIO46 are strapping pins. GPIO0 and GPIO46 set boot mode; GPIO45 sets VDD_SPI voltage (0: 3.3 V, 1: 1.8 V) with default eFuses. GPIO46 = 1 with GPIO0 = 0 is an invalid combination. | [2], [3]                |

## 7. Firmware Requirements

Behaviour the hardware relies on. Each item is required unless marked optional.

| ID   | Requirement                                                                                                                                                        | Reason                                                                                                                                                                     | Source     |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| FW1  | Power sequencing: drive SHDN_BB_3.9 high, wait for PG_+3V3 and PG_-3V3, then assert TRI_STATE_22 / 24. On shutdown, release TRI_STATE before dropping SHDN_BB_3.9. | Oscillators run on +3V3_ANALOG; a high on TRI_STATE (+3V3_DIG GPIO) with the rail off back-powers them through the enable input.                                           | [8], [15]  |
| FW2  | Enable at most one oscillator (TRI_STATE_22 or TRI_STATE_24) at any time.                                                                                          | Both drive the DAC XI input.                                                                                                                                               | [15]       |
| FW3  | Treat PG_+3V3 as valid only while the 3.9 V rail is enabled.                                                                                                       | The LT3042 PG is high impedance in shutdown and reads good while the rail is off.                                                                                          | [16], [18] |
| FW4  | Drive EN_SPEAKER high only $\approx 214\text{ ms}$ after enabling the analog rails; drive it low on a PG fault.                                                    | Rails must be settled before the OPA1622 output is enabled; DC-fault protection.                                                                                           | [16], [18] |
| FW5  | Disable the internal pull-ups on GPIO43 / GPIO44 and disable the UART0 console.                                                                                    | Internal $\approx 45\text{ k}\Omega$ pull-ups against the $10\text{ k}\Omega$ pull-downs leave $\approx 0.6\text{ V}$ idle (thin low margin); console output drives U0TXD. | [2], [8]   |
| FW6  | Debounce all buttons in software.                                                                                                                                  | No hardware debounce fitted.                                                                                                                                               | [8]        |
| FW7  | Run I2C bus recovery (clock out SCL until SDA is released, then STOP) at start-up.                                                                                 | GPIO4 / GPIO5 power-up glitches can present a false START to the DAC and charger.                                                                                          | [2]        |
| FW8  | Keep GPIO1_DAC / GPIO2_DAC low (input, or driven low) until DAC Register 8 reconfigures them.                                                                      | Default DAC mode shuts the DAC down when either pin is high.                                                                                                               | [15]       |
| FW9  | Set ADDR_DAC to its final level before releasing RESETB_DAC.                                                                                                       | DAC I2C address must be stable when it leaves reset.                                                                                                                       | [15]       |
| FW10 | Run the SD card in 1-bit SDMMC mode at up to 40 MHz; treat CD_SD low as card present, debounce it, and mount or unmount on change.                                 | Hardware is 1-bit; CD_SD has a $10\text{ k}\Omega$ pull-up and the switch pulls it low with a card inserted.                                                               | [8], [19]  |
| FW11 | Decode ENC_A / ENC_B with the PCNT peripheral and enable its glitch filter.                                                                                        | Contact chatter up to a few ms remains after the $10\text{ nF}$ RC filter.                                                                                                 | [20]       |
| FW12 | Optional: lower GPIO drive strength on SPICLK / SCL_LCD if EMI or audio coupling appears.                                                                          | Slower edges with no added parts.                                                                                                                                          | Advice     |

## 8. Open Problems

| # | Problem                                                                                   |
|---|-------------------------------------------------------------------------------------------|
| 1 | SELECT node tied to +3V3_DIG in the Interface Buttons block (C18).                        |
| 2 | PG_-3V3 pull-up R222: 10 kΩ exceeds the LT3093 PG low specification (C16).                |
| 3 | Backlight PWM on GPIO48; drive circuit open in Peripherals Rev 9 (C7).                    |
| 4 | Optional: 1 kΩ series on SELECT, VOLUME+ and VOLUME- against the power-up glitch (C17).   |
| 5 | Schematic housekeeping: flash value text reads 64Bit; annotate designators before layout. |

Source: [2], [8], [9], [17].

## 9. Revision Log

| Rev | Date       | Change                                                                                                                                                                                                                                                                                                                                                                                              |
|-----|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0   | 2026-09-29 | Initial issue: ESP32-S3 base chip, 40 MHz crystal, design references, schematic review items.                                                                                                                                                                                                                                                                                                       |
| 1   | 2026-09-29 | Crystal changed to ECS-400-10-37B2-CKY-TR (10 pF). Load capacitors, XTAL_P series inductor and VDD3P3 filter inductor added. Pin default configuration and power-up glitch tables added. Guideline notes added (SPI EMC). Guideline source moved to the online ESP32-S3 Hardware Design Guidelines. Open problems 1, 8, 9 updated; 11 and 12 added. All entries cited.                              |
| 2   | 2026-09-29 | Crystal load capacitors changed to 12 pF GJM1555C1H120FB01D (#1-122); $C_{L,eff}$ 9 to 11 pF recorded. Open problem 1 (load capacitor value) closed; open problems renumbered.                                                                                                                                                                                                                      |
| 3   | 2026-09-29 | GPIO assignment table (Section 3.3) from the current schematic sheets, with recommended changes and comments. Closed: LCD off flash pins, CHIP_PU RC, VDDA supply, I2C crossing, strapping (EN1_BQ removed, CE_BQ on GPIO46, CS_LCD pull-down on GPIO45), power-up glitch review. Added: boot access, SHDN_BB_3.9 pull-down, U0TXD series resistor, GPIO budget, button polarity, flash-pin stubs.  |
| 4   | 2026-09-29 | Flash selected: GigaDevice GD25B64EWIGR (8 MB, quad SPI, WSON-8). SHDN_BB_3.9 pull-down part #1-123 (1 MΩ). BOOT and RESET buttons: C&K; PTS810 (reused). Flash pin mapping in the GPIO table. Open problems updated.                                                                                                                                                                               |
| 5   | 2026-09-30 | Flash added to library (#1-124, SON127P600X500X80-9N). Package dimensions and pinout from the GD25Q64E datasheet recorded; land pattern review results recorded. Open problem 1 narrowed.                                                                                                                                                                                                           |
| 6   | 2026-09-30 | Reviewed against DAP-MCU-3 of 2026-09-30. Closed: flash datasheet and footprint, PSRAM (not used), SHDN_BB_3.9 pull-down, flash wiring, supply decoupling. GPIO table updated (flash nets, GPIO35 pull-down, GPIO0 BOOT wiring, GPIO26 free). Added: BOOT switch rail, SPICLK EMC footprint, housekeeping; C15.                                                                                     |
| 7   | 2026-10-01 | Firmware Requirements section added (FW1 to FW10). Closed: boot access, DOWN series resistor, SPICLK series resistor, button polarity (active high kept). Added: PG_-3V3 pull-up (C16), button power-up glitch (C17), Peripherals polarity update.                                                                                                                                                  |
| 8   | 2026-10-01 | SD and encoder assigned: CLK_SD GPIO47, CMD_SD GPIO41, DAT0_SD GPIO42, CD_SD GPIO26, ENC_A GPIO16, ENC_B GPIO15; SELECT moved to GPIO12 (encoder switch), TRI_STATE_24 to GPIO3. Storage requirement set to 1-bit SDMMC. FW10 and FW11 added; optional drive-strength item renumbered FW12. Closed: GPIO budget, microSD selection, Peripherals polarity, C2. Added: SELECT tied to +3V3_DIG (C18). |

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## 10. Sources

1. [ESP32-S3 Hardware Design Guidelines \(Espressif\)](#)
2. [ESP32-S3 Hardware Design Guidelines, Schematic Checklist \(Espressif\)](#)
3. [ESP32-S3 Series Datasheet \(Espressif\)](#)
4. [ECS-400-10-37B2-CKY-TR Product Page \(DigiKey\)](#)
5. [ECS-400-10-37B2-CKY-TR Product Page \(ECS Inc.\)](#)
6. [GJM1555C1H120FB01D Product Page \(DigiKey\)](#)
7. [MHQ1005075HA Series Inductors \(TDK, via DigiKey\)](#)
8. DAP schematic sheets DAP-MCU-3 and DAP-Peripherals-4 (2026-10-01, latest), DAP-Audio-1, DAP-Power-2 (author, 2026-09-29)
9. Peripherals: Working Document, Rev 9
10. LTC3440 datasheet, SHDN/SS pull-down value (as reported by the author)
11. [GD25B64EWIGR Product Listing \(Dasenic\)](#)
12. RE0402BRE071ML, author library entry #1-123 (DigiKey 13-RE0402BRE071MLCT-ND)
13. [GD25Q64E Datasheet \(GigaDevice\), Sections 2 and 10.6](#)
14. [GD25B64C Product Page \(GigaDevice\)](#)
15. DAC Clocking and I2S Interface (ES9038Q2M): Working Document, Rev 9 (project file)
16. Positive LDO (LT3042): Working Document, Rev 6 (project file)
17. Negative LDO (LT3093): Working Document, Rev 9 (project file)
18. Output Stage (OPA1622): Working Document, Rev 4 (project file)
19. [MSD-1-A Product Page \(Same Sky\)](#)
20. [EC12D1564404 Product Page \(Alps Alpine\)](#)